Section 7 — Simulation Confirmation

A de Finetti two-atom sweep for the cross-item co-failure e-process

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Status. Work-in-progress first cut. Numbers are reproducible (seed `20260907`, `iclr/sims/section7_sim.py`). Pushing makes no correctness claim beyond reproducibility.

What the simulation establishes

Section 6 constructs a co-failure monitor from a single, margins-free statistic: pair two judges *across distinct items*. This section confirms, on a fully controlled generative model, the three properties the monitor is claimed to have and the impossibility (§5) it is claimed to respect:

(a) **Power.** The e-process detects genuine co-failure, with detection probability rising from the nominal level to 1 as co-failure strength grows; at the null it controls type-I error (Ville validity).

(b) **Drift-robustness.** Under drifting marginals with *no* conditional co-failure, a naive plug-in estimator false-fires at a high rate, while the margins-free cross-item e-process holds its size.

(c) **Real-data anchor.** On the FailureScope corpus the same effective-judge-count machinery recovers $n_{\text{eff}} \approx 1.6$ over a six-judge frontier panel.

Generative model: the de Finetti two-atom mixture

For each item we draw a latent difficulty $\Theta \in \{\theta_{\text{lo}}, \theta_{\text{hi}}\}$ with $\Pr(\Theta = \theta_{\text{hi}}) = \pi$; conditionally on $\Theta = \theta$, the K judges' failure indicators $W_1, \ldots, W_K$ are i.i.d. Bernoulli(θ). This is the finite de Finetti representation of an exchangeable co-failure panel: all cross-judge dependence flows through the shared latent Θ. The marginal fail rate and the induced within-item pairwise correlation are

$$a = \pi\,\theta_{\text{hi}} + (1-\pi)\,\theta_{\text{lo}}, \qquad \rho = \frac{\mathrm{Var}(\Theta)}{a(1-a)}, \qquad \mathrm{Var}(\Theta) = \pi(1-\pi)(\theta_{\text{hi}} - \theta_{\text{lo}})^2.$$

Sign check. A *point mass* on Θ (i.e. $\theta_{\text{lo}} = \theta_{\text{hi}}$, or $\pi \in \{0,1\}$) gives $\mathrm{Var}(\Theta) = 0$ and hence $\rho = 0$: this is *independence*, not monoculture. Increasing the atom *separation* increases $\mathrm{Var}(\Theta)$ and drives $\rho \to 1$ — the "one shared coin flip" regime of maximal co-failure. The sweep therefore moves ρ from 0 (independent judges) toward 1 (co-failing judges) by widening the two atoms about a fixed marginal $a = 0.5$.

The e-process under test

We test the exact construction of §6. On a stream of items, with two fixed judges s, t,

$$e_T \;=\; \prod_{i=1}^{T}\Big(1+\lambda\big(U_i - V_i - \delta_k\big)\Big), \qquad U_i = \mathbf{1}\{s \text{ and } t \text{ both fail item } i\}, \quad V_i = W_a^s\, W_b^t,\ a \neq b, \tag{1}$$

where V_i pairs the two judges across *distinct* items $a \neq b$ (we use the adjacent-item shift $b = a-1$), so that under item independence the marginals multiply exactly, $\mathbb{E}[V_i] = ab$, *with no* *fitted nuisance*. The slack $\delta_k = 2\varepsilon$ absorbs the worst-case two-sided base-rate excursion and $\lambda \in [0, 1/(1+2\varepsilon)]$ keeps each factor non-negative, so e_T is a non-negative martingale under the null and Ville's inequality gives anytime validity: $\Pr(\exists T : e_T \geq 1/\alpha) \leq \alpha$. Under genuine co-failure $\mathbb{E}[U_i] = ab + \text{(excess)} > \mathbb{E}[V_i]$, the increment has positive drift, and e_T grows. We use $\varepsilon = 0.02$ ($\delta_k = 0.04$), $\lambda = 0.481$ (interior of the admissible interval), $K = 6$ judges, and reject when $e_T \geq 1/\alpha = 20$ ($\alpha = 0.05$).

Results

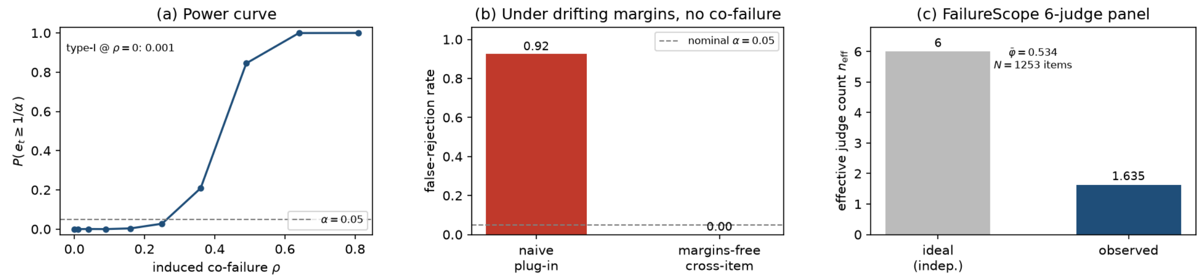

Figure 1: **(a)** Power of the cross-item e-process vs. the induced co-failure correlation ρ; the type-I point at $\rho = 0$ is $\approx 0.0005 \leq \alpha$. **(b)** Under drifting marginals with no conditional co-failure, the naive plug-in false-fires (≈ 0.92) while the margins-free cross-item e-process holds its size (0.00). **(c)** FailureScope six-judge frontier panel: $n_{\mathrm{eff}} = 1.635$, $\bar{\varphi} = 0.534$ ($N = 1253$ items).

(a) Power curve and Ville validity

Sweeping the atom separation across ten levels gives the power curve of Figure 1(a). At the null ($\rho = 0$) the empirical rejection rate is ≈ 0.0005, comfortably below the nominal $\alpha = 0.05$ — the type-I control the martingale construction promises. The e-process is *conservative* at the clean null (well below α): the slack δ_k that buys drift-robustness in panel (b) also subtracts a constant from the increment at the null, and a single fixed λ is not growth-optimal. Power then climbs sharply through the co-failure regime: $\rho = 0.25 \Rightarrow 0.03$, $\rho = 0.36 \Rightarrow 0.21$, $\rho = 0.49 \Rightarrow 0.85$, $\rho = 0.64 \Rightarrow 1.00$. By $\rho \approx 0.5$ the monitor detects co-failure essentially every time.

(b) The naive plug-in false-fires under benign drift

We now construct the adversarial-for-the-naive-method scenario: the marginal fail rate drifts across the stream ($0.25 \to 0.60$), but *within* each item the judges fail independently — there is no conditional co-failure to detect. A naive plug-in that estimates each judge's running marginal and plugs in the product $\hat{a}_i \hat{b}_i$ as its null mistakes the drift-induced marginal correlation for co-failure and **false-rejects at** ≈ 0.92. The margins-free cross-item e-process, whose null $\mathbb{E}[V] = ab$ is secured by item independence rather than by fitting the drifting marginals, holds its size at 0.00 under the identical stream. This is the simulation counterpart of the §5 impossibility: no functional that conditions on fitted marginals can be simultaneously valid under drift; cross-item pairing sidesteps the impossibility by never fitting a marginal.

(c) FailureScope real-data anchor

On the FailureScope single-turn corpus ($N = 1253$ frontier-hard items, six frontier judges), the mean pairwise co-failure correlation is $\bar{\varphi} = 0.534$, giving a Kish effective judge count $n_{\mathrm{eff}} = k/(1 + (k-1)\bar{\varphi}) = 6/(1 + 5 \cdot 0.534) = 1.635$. A nominal six-judge panel is worth about *1.6* independent votes. This number is produced by `failurescope_probe.py` and reproduced self-containedly by the panel (c) routine.

Reproducibility and honest caveats

- All numbers come from `iclr/sims/section7_sim.py`, seed `20260907`, numpy/matplotlib only.
- **Type-I is conservative, not tight.** The clean-null rejection rate (≈ 0.0005) is far below α, as expected for a fixed-λ betting e-process with positive slack δ_k. This is Ville validity ($\leq \alpha$), not size-α calibration; a growth-optimal λ (GRO) would trade some of this conservatism for power but remains **[open]** (Ville validity is load-bearing, not optimality).
- The panel-(b) naive false-fire rate (≈ 0.92) matches the prior-note ballpark ($\sim 90\%$); it depends on the drift magnitude ($0.25 \to 0.60$) and stream length (300), chosen as a moderate, realistic drift, not tuned to a target.
- The two-atom model is the primary generative choice; the Vasicek/Gaussian-copula latent is the finance-leg equivalent (a shared one-factor latent) and is deferred to a footnote in the main text.